

CDTRS V2

Backend, Database & API Integration Specification

Revised Comprehensive Edition — Consolidated Requirements

Purpose

This revised specification preserves the original CDTRS V2 architecture and incorporates the additional backend requirements identified during the latest workflow and UI review. It is intended to be given to the backend developer before implementation starts.

Core principle: CDTRS remains document-centric. One canonical document_id must survive the complete lifecycle from intake through closure. The new requirements extend this model rather than replacing it.

Decision	Baseline
Frontend	PySide6 / Qt Widgets / service layer / centralized API client
Backend	FastAPI / SQLAlchemy / Pydantic / authentication / authorization
Database	PostgreSQL; never connected directly by frontend
Roles	DS, DIRECTOR, HOD, EMPLOYEE
Workflow	DS → Director → DS → HOD/Employee → Progress → DS → Director → DS → Closed
Storage	Backend-managed files; database stores storage references
OCR	Asynchronous OCR + structured extraction + verification
Routing	Suggestion → DS review/edit → confirmed route
Visibility	Backend-enforced role, department and employee scope
Updates	Event-driven/live updates; no manual Refresh dependency

1. What This Revision Adds

The original specification already establishes the core database entities, workflow, API direction, authorization rules, attachment model, document history and frontend/backend boundary. This edition adds the operational details that became necessary during system testing and review.

Area	New/clarified requirement
Mail intake	Track incoming message separately from the document and preserve sender, subject, timestamp and message identifier.
Mail attachments	Store each incoming attachment with source-message provenance and prevent duplicate ingestion.
File storage	Define a backend storage abstraction and never expose physical paths to the frontend.
OCR	Add OCR job lifecycle, full-text artifact, structured extracted fields and extraction provenance.
OCR verification	Never let re-analysis overwrite DS-verified values silently.
Routing intelligence	Store department/employee suggestions, confidence, reason and source separately from confirmed routes.
Director remark extraction	Surface explicit department/employee instructions as a highlighted suggestion for DS.
Remarks	Keep latest remark on document for fast display and retain complete remark history separately.
Visibility	Hard-enforce HOD department scope and employee personal scope in backend queries.
Reminders	If employee is assigned, remind employee; otherwise if department exists, remind department HOD.
Live updates	Use backend events/WebSocket/SSE or equivalent so relevant screens update without Refresh buttons.
Reliability	Use transactions, idempotency and optimistic concurrency for workflow operations.
Testing	Seed two HODs from different departments and test cross-department isolation.

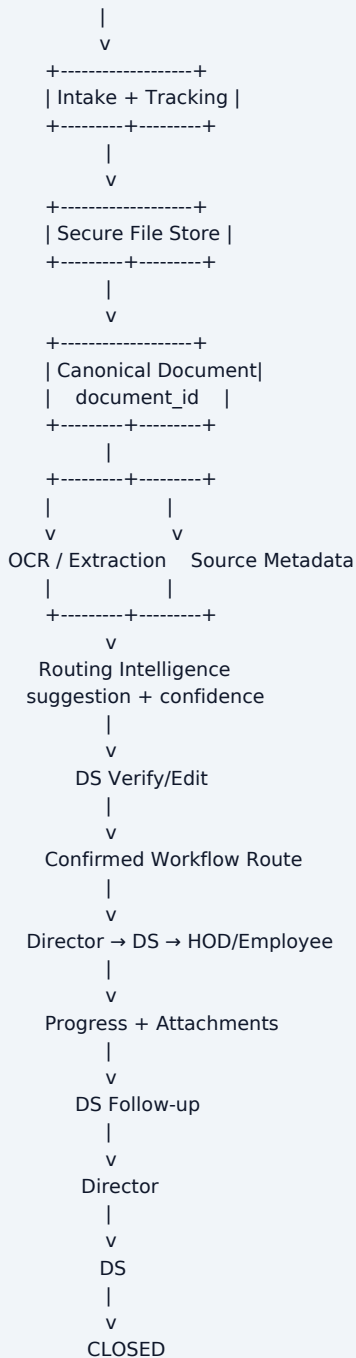
2. Non-Negotiable Architecture Principles

- One document: do not create separate Director/HOD/Employee document records.
- Routing is not assignment: DS routes; HOD assigns employee.
- Suggestion is not routing: OCR/AI may recommend, but DS confirms.
- Backend is authoritative: frontend hiding/filtering is not security.
- Original files are retained: OCR output is derived data, not a replacement for the source file.
- History is retained: remarks, progress, routes, attachments and workflow events are not overwritten.
- Closed means closed: normal workflow mutations are rejected after closure.

3. Target Architecture

Incoming Sources

Outlook / Government Mail / Manual Upload



Backend modules should be separated so that ingestion, storage, OCR, routing, workflow, authorization, notifications and events can be tested independently.

Module	Responsibility
Authentication	Login, identity, token/session handling
Authorization	Role, department, document and attachment access
Intake	Mail synchronization, incoming message records, attachment ingestion
Document service	Canonical document metadata and lifecycle
Storage service	Upload, storage, retrieval and secure download
OCR service	OCR jobs, extracted text, fields, confidence and retries
Routing service	Routing suggestions and evidence

Module	Responsibility
Workflow service	State validation and transitions
Remark service	Current remarks + remark history
Assignment service	HOD → Employee assignments
Progress service	Multiple employee updates and supporting files
Notification service	Notifications and deadline reminders
Event service	Live update events for connected clients
Audit service	Security/administrative audit records

4. Canonical Workflow

#	Actor	Action	Backend result
1	DS	Receive/process intake	Source + file + canonical document
2	DS	Verify extracted metadata	Verified values saved
3	DS	Route to Director	Route + history + notification
4	Director	Review document	Authorized file/document access
5	Director	Save/edit remark	Latest remark + remark history
6	Director	Return to DS	DS stage + history + notification
7	DS	Review Director instruction	Routing suggestion may be generated
8	DS	Edit/confirm route	Confirmed HOD or employee route
9	HOD	Optional remark / assign employee	Assignment + notification
10	Employee	Work and submit progress	Progress retained
11	HOD/DS	Monitor	Role-scoped progress visibility
12	DS	Forward relevant progress	Follow-up to Director, same document_id
13	Director	Final remark	Remark retained + return to DS
14	DS	Close	CLOSED + closure timestamp

4.1 Director remark and return are separate actions. Saving a remark must not internally perform Return to DS. The UI may provide a convenient combined workflow action, but the backend concepts remain independent.

4.2 HOD remark and assignment are also separate. HOD must not be forced to enter a second/duplicate remark merely to assign an employee.

4.3 The Director cannot route to HOD/Employee. If the Director writes an explicit routing instruction, the system detects it and surfaces it to DS; DS remains the authority who confirms the route.

5. Incoming Mail / Outlook Integration

A mail item and its attachment are different source objects. The backend must preserve the relationship so that CDTRS can answer: who sent this, what was the subject, when was it received, which attachment became the document, and where the original file is stored?

```
Mail system
|
+-- external_message_id
+-- sender_name
+-- sender_email
+-- subject
+-- received_at
+-- body/reference
+-- attachments
    |
    v
    CDTRS Intake
    |
    +--> incoming_message
    +--> attachment storage
    +--> canonical document
    +--> OCR queue
```

5.1 `incoming\_messages`

Field	Purpose
id	Internal source-message ID
source_type	OUTLOOK / GOVERNMENT_MAIL / OTHER_APPROVED_SOURCE
external_message_id	Original message identifier; used for de-duplication
sender_name	Sender display name
sender_email	Sender address
subject	Original subject
received_at	Original received time
body_reference	Stored/reference representation if retained
has_attachments	Boolean
processing_status	NEW / PROCESSING / PROCESSED / FAILED / IGNORED
created_at	CDTRS ingestion timestamp

5.2 Intake rules

- Repeated synchronization of the same external message must not create duplicate CDTRS documents.
- Every incoming attachment gets a storage record and retains original filename and MIME type.
- Multiple attachments must be supported.
- The system should identify the primary document attachment without deleting supporting attachments.
- A message with no usable document may remain an intake item for DS review.
- Manual upload should use the same canonical storage/document pipeline as mail intake.

6. File Storage and Attachments

The database should store references to files, not large binary contents. The frontend must use backend access/download endpoints and must never assume a Windows/Linux physical storage path.

Logical storage
storage/
incoming/
documents/
progress/
processed/
temporary/

Database:
storage_key
original_filename
mime_type
file_size
checksum
attachment_type
created_at

6.1 Attachment types

Type	Meaning
ORIGINAL	Canonical document uploaded directly to CDTRS
EMAIL_ATTACHMENT	Original file received from incoming mail
SUPPORTING_DOCUMENT	Additional workflow document
PROGRESS_ATTACHMENT	File submitted with a particular employee progress update

6.2 Recommended `attachments` fields

Field	Purpose
id	Primary key
document_id	Canonical document
progress_update_id	Nullable; links progress attachment to its progress update
uploaded_by_user_id	Uploader
file_name	Original filename
storage_key	Backend storage reference
file_type	MIME/type
file_size	Size
checksum	Integrity/de-duplication
attachment_type	Original/email/supporting/progress
source_message_id	Nullable originating mail item
created_at	Timestamp

6.3 Secure download

GET /attachments/{attachment_id}/download

Backend:

1. Authenticate user
2. Resolve attachment → document
3. Authorize user against document
4. Resolve storage_key
5. Stream file
6. Never expose raw filesystem path

7. OCR and Extraction Pipeline

OCR is a backend processing pipeline. The full extracted text may be many pages long and should not be dumped into the normal Document Information panel. The UI should show concise structured fields and provide an appropriate document/OCR viewer only when needed.

```
Stored original
|
v
OCR = PENDING
|
v
OCR = PROCESSING
|
+--> full extracted text
+--> page text if supported
+--> structured fields
+--> routing clues
|
v
OCR = COMPLETED / FAILED
```

OCR status	Meaning
PENDING	Waiting for worker
PROCESSING	OCR/extraction running
COMPLETED	Extraction available
FAILED	Processing failed; retry/re-analysis permitted

7.1 Suggested `document\_ocr` fields

Field	Purpose
id	OCR record
document_id	Canonical document
extracted_text	Full OCR text artifact
ocr_status	Processing state
ocr_engine	Engine/version
confidence	Overall or normalized confidence
processed_at	Completion timestamp
error_message	Failure information if applicable

7.2 Suggested `document\_extracted\_fields`

Field	Purpose
id	Record ID
document_id	Canonical document
field_name	TITLE, REFERENCE, DEPARTMENT, EMPLOYEE, etc.
extracted_value	Raw extracted value
confidence	Field confidence
source_page	Page number where found
source_text	Supporting snippet if available
verified_value	DS-verified value
verified_by / verified_at	Verification provenance

7.3 Re-analysis rule: a new OCR result may replace an unverified suggestion, but it must not silently replace a value already verified or manually edited by DS.

8. Routing Intelligence

Routing intelligence uses extracted document content, source metadata and Director remarks to recommend a destination. It is advisory until DS confirms it.

Field	Purpose
suggested_department_id	Recommended department
suggested_employee_id	Recommended employee if resolved
routing_confidence	Confidence score/band
routing_reason	Human-readable explanation
routing_source	DOCUMENT_CONTENT / DIRECTOR_REMARK / SOURCE_METADATA / MANUAL
generated_at	Suggestion timestamp
confirmed_by	DS who confirmed
confirmed_at	Confirmation timestamp

POST /documents/{id}/analyze-routing
-> create/update suggestion only

GET /documents/{id}/routing-suggestion
-> return current suggestion

POST /documents/{id}/route
-> confirm actual workflow route

8.1 Explicit Director instruction

Director remark:
"Assign this to Rahul Sharma, Finance."

Detected:
employee = Rahul Sharma
department = Finance
confidence = 91%
source = DIRECTOR_REMARK

DS UI:
WARNING: Explicit routing instruction detected
Suggested Department: Finance
Suggested Employee: Rahul Sharma
[Edit] [Confirm Routing]

No route changes until DS confirms.

8.2 If no employee is explicitly identified, DS routes to the appropriate HOD. If an employee is explicitly identified and can be resolved to an active employee/user, direct DS → Employee routing is allowed.

9. Consolidated Database Model

Core entities from the original specification remain: users, departments, employees, documents, document_routes, work_assignments, progress_updates, attachments, workflow_history, audit_logs and notifications. The following additions/clarifications are required.

9.1 `documents`

Field	Purpose
doc_id	Primary key
reference_no	Human-readable CDTRS ID
title	Verified title
description	General description
received_date	Received date
deadline	Target deadline
source	Sender/source organization
mode	Ingestion mode/source type
format / mime_type	Original file format
priority	HIGH / MEDIUM / LOW
status	User-facing lifecycle status
current_stage	DS / DIRECTOR / HOD / EMPLOYEE / CLOSED
current_owner_id	Current responsible user where applicable
target_department_id	Relevant department
created_by	Creator/receiver
director_remark	Latest Director remark
hod_remark	Latest HOD remark
source_message_id	Nullable incoming-message link
ocr_status	Current OCR state
version	Optimistic concurrency version
created_at / updated_at / closed_at	Timestamps

9.2 `document\_remarks`

Field	Purpose
id	Primary key
document_id	Canonical document
author_user_id	Author
role	Author role
remark_text	Remark
remark_type	DIRECTOR / HOD / other approved type
created_at	Created time
updated_at	Last edit time

Keep latest Director/HOD values on `documents` for quick display, while `document\_remarks` preserves the full history of edits.

10. Role and Department Visibility

Visibility must be enforced in backend queries and service methods, not merely by filtering a table in PySide6.

Role	Permitted visibility
DS	Documents under DS responsibility across intake, Director review, HOD processing, employee progress, follow-up and closure.
Director	Only documents sent to Director by DS and relevant follow-up of those documents.
HOD	Only documents routed to that HOD/department, including permitted history, assignments, progress and attachments.
Employee	Only documents directly routed to that employee or assigned to that employee by HOD.

10.1 Two-HOD test requirement

Finance Department
HOD Finance
Rahul Sharma
other Finance employees

Procurement Department
HOD Procurement
Priya Verma
other Procurement employees

Test	Expected
Finance HOD opens inbox	Finance documents only
Finance HOD requests Procurement document	Denied
Procurement HOD opens history	Procurement documents only
Finance employee opens tasks	Own permitted tasks only
Employee requests another employee's attachment	Denied
Director opens department-only document not sent to Director	Denied

The backend may return 403 or intentionally use 404 for inaccessible resources depending on the chosen resource-hiding policy, but it must never leak the protected document.

11. Remarks, History and Workflow Audit

Workflow history remains document-centric and user-visible. System/security audit logs remain separate. The backend should retain all workflow events even if the UI presents history compactly.

Workflow history	Audit log
Document-specific	System/security/administrative
DS routed to Director	Login/logout
Director saved remark	Authorization failure
HOD assigned employee	Administrative configuration
Employee submitted progress	Security event
Document closed	System-level activity

Required workflow history fields: id, document_id, performed_by_user_id, action, from_role, to_role, details, created_at.

The document viewer should not reserve a large permanent area for the entire workflow table. History can be opened/expanded for the document, while the dedicated History/Audit screen remains the system-wide document-centric view.

11.1 Remark editing rules

- Director can edit/save Director remark while the document is in a Director-appropriate stage.
- HOD can edit/save HOD remark for department documents.
- Employee does not overwrite HOD/Director remarks.
- Saving a remark creates/updates the latest value and records the change in remark history.
- Saving a remark is not itself a route/assignment transition.

12. Notifications and Reminder Logic

The notification entity from the original specification remains valid. The recipient-selection logic is expanded to cover unassigned departmental work.

```
if active employee assignment exists:
    recipient = assigned employee

elif target department exists:
    recipient = active HOD of target department

else:
    recipient = configured DS fallback
```

This means a department-routed document with no staff assignment generates an action reminder for the HOD. Once staff is assigned, subsequent employee-specific reminders can target the assigned employee.

Reminder field	Purpose
document_id	Document needing action
recipient_user_id	Resolved recipient
reason	DUE_SOON / OVERDUE / ACTION_REQUIRED
due_at	Trigger time/deadline
sent_at	Sent time
is_read	Read state
deduplication_key	Prevent repeated identical reminders

12.1 Live updates

Manual Refresh buttons should not be required for normal workflow changes. Use WebSocket, Server-Sent Events, or another suitable event mechanism so the relevant client can update after document routing, remark save, assignment, progress, attachment upload, closure and notification creation.

Event	Typical affected views
DOCUMENT_CREATED	DS inbox/dashboard
DOCUMENT_ROUTED	DS/HOD/Employee queues
REMARK_UPDATED	Relevant document viewers
ASSIGNMENT_CREATED	HOD + employee
PROGRESS_SUBMITTED	Employee/HOD/DS
ATTACHMENT_ADDED	Document/progress view
DOCUMENT_CLOSED	DS lists/dashboard
NOTIFICATION_CREATED	Notification bell/unread count

13. API Contract — Consolidated

Method	Endpoint	Purpose
POST	/auth/login	Authenticate
GET	/auth/me	Current user
POST	/auth/logout	Logout/session invalidation
GET	/documents	Authorized listing
GET	/documents/inbox	Role-scoped inbox
POST	/documents	Create document/manual intake
GET	/documents/{id}	Authorized document
POST	/documents/{id}/route	Confirm route
PUT	/documents/{id}/director-remark	Save/edit Director remark
POST	/documents/{id}/return-to-ds	Director return
PUT	/documents/{id}/hod-remark	Save/edit HOD remark
POST	/documents/{id}/assign	HOD assignment
POST	/documents/{id}/progress	Employee progress
GET	/documents/{id}/progress	Progress list
POST	/documents/{id}/attachments	Upload attachment
GET	/documents/{id}/attachments	List attachments
GET	/attachments/{id}/download	Authorized download
GET	/documents/{id}/history	Document history
POST	/documents/{id}/follow-up	DS follow-up to Director
POST	/documents/{id}/close	Close document
GET	/notifications	Notifications
GET	/notifications/unread	Unread notifications
PATCH	/notifications/{id}/read	Read notification
GET	/intake	Intake list
POST	/intake/manual-upload	Manual intake
POST	/intake/{id}/process	Process intake
POST	/documents/{id}/process-ocr	Start/restart OCR
GET	/documents/{id}/ocr	OCR status/results
POST	/documents/{id}/reanalyze	Re-run approved analysis
POST	/documents/{id}/analyze-routing	Generate routing suggestion
GET	/documents/{id}/routing-suggestion	Get routing suggestion
GET	/dashboard	Role-appropriate dashboard

13.1 Standard errors

401 = not authenticated
 403 = authenticated but unauthorized
 404 = not found or intentionally hidden
 409 = workflow/concurrency conflict
 422 = request validation failure
 500 = unexpected server error

14. Authentication and Backend Authorization

The final system must not use hardcoded credentials in frontend code. Authentication, authorization and token/session handling are backend responsibilities.

- Store password hashes only.
- Resolve current authenticated user on every protected request.
- Apply role checks server-side.
- Apply department checks for HOD operations.
- Apply employee ownership/assignment checks for employee operations.
- Re-check authorization before every file download.
- Never rely on a disabled/hidden UI button as permission enforcement.

15. Transactions, Idempotency and Concurrency

Multi-record workflow actions must be atomic.

Operation	Atomic side effects
DS confirms route	document state + route + workflow history + notification + event
Director return	document state + history + notification + event
HOD assignment	assignment + owner/stage + history + notification + event
Progress	progress update + attachment links + history + notifications/events
DS follow-up	follow-up route + history + Director notification
Closure	closed state + closed_at + history + notification/event

Use an idempotency/deduplication strategy for mail ingestion and other operations that may be retried. Use a document version number or equivalent optimistic concurrency mechanism. A stale update should return 409 Conflict instead of silently overwriting a newer state.

16. Background Processing

OCR and mail synchronization can take longer than a normal UI request. Use background workers/jobs and expose status to the frontend.

```
POST /documents/{id}/process-ocr
|
+--> 202 Accepted
|
v
ocr_status = PROCESSING
|
v
worker completes
|
+--> fields saved
+--> routing suggestion optionally generated
+--> event published
```

17. Closure Rules

After the final Director review, the Director returns the document to DS. DS is the role that closes the document.

On closure: status becomes CLOSED, closed_at is populated, a workflow event is recorded, and normal route/assign/progress workflow operations are rejected. Existing documents, history, remarks, progress and attachments remain viewable according to authorization.

18. Testing and Required Seed Data

Suite	Minimum test
Authentication	Valid/invalid login, session/token behavior
Authorization	DS/Director/HOD/Employee permissions
Department isolation	Two HODs cannot cross-access
Employee isolation	Employee cannot see another employee's tasks/files
Workflow	Only valid state transitions
Routing	HOD vs direct employee rules
Routing suggestion	Suggestion does not itself route
OCR	Pending/processing/completed/failed/retry
Verification	Re-analysis does not overwrite verified fields
Mail	Duplicate external message does not duplicate document
Attachments	Original/progress linkage + download authorization
Reminders	Employee first; HOD fallback when unassigned
Live events	Relevant clients update without Refresh
Concurrency	Stale write returns 409
Closure	Closed document rejects normal workflow mutation

Required test users should include DS, Director, at least two HODs from different departments, and multiple employees across those departments.

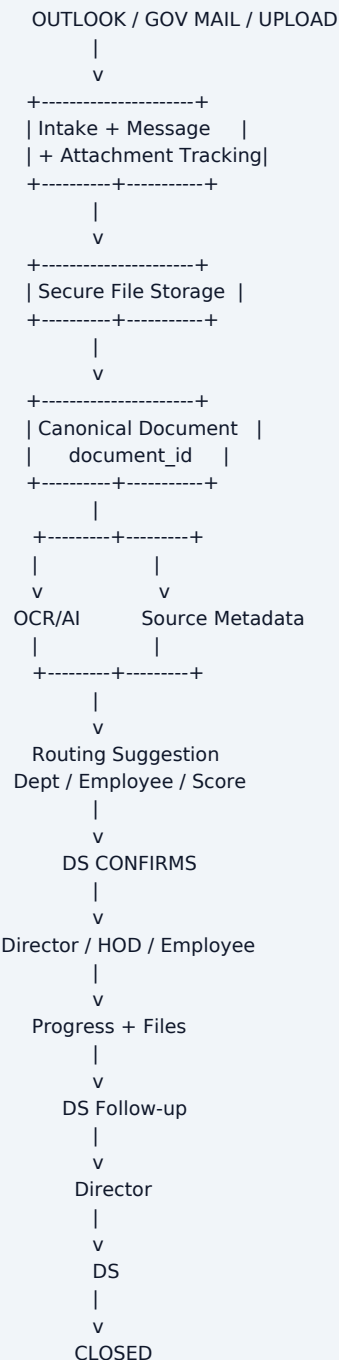
19. Recommended Implementation Order

- 1 Foundation: FastAPI configuration, PostgreSQL, SQLAlchemy, migrations and logging.
- 2 Authentication: users, password hashing, login and current-user handling.
- 3 Authorization: reusable role/department/document access checks.
- 4 Core database: departments, employees, users, documents, routes, assignments, progress, attachments, workflow history, notifications and audit logs.
- 5 Storage: storage abstraction and secure upload/download.
- 6 Manual intake: establish canonical document creation first.
- 7 Mail intake: incoming message model and mail integration adapter.
- 8 OCR: asynchronous processing and structured extraction.
- 9 Verification: preserve extracted and verified values separately.
- 10 Routing: suggestion, confidence, source/provenance and DS confirmation.
- 11 Workflow: Director, HOD, employee, follow-up and closure services.
- 12 Notifications: action reminders and recipient fallback.
- 13 Live events: event publication/subscription.
- 14 Reliability: transactions, idempotency and optimistic concurrency.
- 15 Testing/deployment: multi-HOD isolation, end-to-end lifecycle and reachable FastAPI environment.

20. Final Acceptance Checklist

Requirement	Acceptance condition
Canonical document	Same document_id through entire lifecycle
Mail ingestion	Message + attachment provenance retained
Storage	No raw filesystem paths exposed to frontend
OCR	Multi-page/scanned files process asynchronously
Extraction	Structured fields available without dumping full OCR into Document Information
Verification	Manual/verified values survive re-analysis
Routing intelligence	Suggestion includes confidence/source and requires DS confirmation
Director instruction	Explicit department/employee instruction is highlighted to DS
HOD isolation	HOD sees only own department's permitted documents/history
Employee isolation	Employee sees only own permitted tasks/files
Remarks	Latest values easy to display; history retained
Assignment	HOD assignment independent of remark
Attachments	Original and progress attachments are linked and permission-checked
Reminders	Assigned employee gets reminder; otherwise department HOD
Live updates	Normal changes appear without manual Refresh
Transactions	Related workflow side effects are atomic
Concurrency	Stale updates safely rejected
Closure	Closed document rejects normal workflow mutations
API	Swagger/OpenAPI and test accounts available
Regression	All role/workflow/ingestion tests pass

21. Final Target Architecture — Operational View



Every important change:
transaction
+ workflow history
+ notification
+ live event
+ authorization checks

Final instruction to backend developer: use this document as the implementation baseline before coding the API/database layer. Do not build isolated screens or endpoints without first enforcing the canonical document model, role/department authorization, workflow state rules, storage contract and transaction boundaries.

The original V2 specification remains the architectural foundation. This revised edition adds the requirements discovered after functional testing so the backend can support the actual CDTRS workflow rather than only the earlier UI/mock workflow.

End of Specification